

Datacenters vs Traditional HPC vs AI

What makes AI computing so special?

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- Traditional HPC: large-scale computing for scientific research
 - Computational fluid dynamics or genome sequencing/assembly
 - Large jobs, few users, long-running
- Enterprise datacenters: computing, storage, and access for businesses
 - Email, web browsing, cloud services
 - Small jobs, many users, short-running
- AI datacenters: large-scale computing for AI
 - AI training and inference (e.g. LLMs, image generation, etc.)
 - 'Large' jobs, many users, short/medium-running

These definitions and delineations are not clear-cut.
There is some overlap and convergence amongst these categories.

**What task do you need to perform,
and what does it take to do that?**

Traditional HPC

What is Traditional HPC?

- 'Large' scale systems (used to be the 'largest' synchronous computing scale)
- 'Supercomputers' - TOP500 list
- High-bandwidth, low-latency interconnects
- Shared filesystem
- Becoming dominated by GPU-based systems (exception: Fugaku)



- 'Datacenters': What hosts our data, webserver, microservices, video streaming, etc.
- Smaller jobs - not much computing
- Emphasized reliability, scalability, latency, and cost-effectiveness
- CPU-dominated, but some HW-accelerated tasks (e.g. YouTube)

- Throughput-focused: AI factories.
 - "Tokens are commodities" - Jensen Huang (paraphrased)
- Dominated by HW accelerators (GPUs, TPUs, etc.)
 - Most AI models rely on matrix-matrix multiplications
- Large jobs - Trillion-parameter AI models
- High-performance interconnects optimizing collective communication for training
- **DOMINATING** the computing market, supply chains, academic discussions, etc.

AI Datacenters



Microsoft Fairwater
(Hyperscale AI datacenter)



Meta's 'Rapid deployment structures'
(Image source: DataCenterDynamics)

- Flop Density: *Exaflop/s* per rack! (MatMuls flip lots of transistors)
- Power Density: Approaching 1 MW per rack!
- Thermal Density: More power over slightly more space - More cooling required!
 - Liquid cooling (now)
 - Sub-ambient liquid cooling (future?)

When hardware fails or underperforms, what happens? What are the consequences?

- Datacenters: Degraded service or failover to backups
- HPC and AI: Jobs likely fail or slow to a crawl
- AI Training: The scale is so large, failure is always imminent.
 - Meta's Llama 3 paper - good section on this
 - Straggler (underperforming) nodes have a large impact
 - How much to checkpoint vs accept loss of progress?

HPC used to dominate the high-performance computing market.

Not anymore.

"HPC is a rounding error in the era of AI"

- Exascale computing project (HPC): Limit power to ≈ 20 MW
- AI Datacenters: Multi-GW and only limited by energy supply
 - Build nuclear plants
 - Make a datacenter constellation of satellites